

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Even Semester)
Innovative Practices Description

UNIT II CELLULAR ARCHITECTURE

Degree, Semester& Branch: VI Semester B.E. ECE A

Course Code & Title: EC8652 Wireless Communication

Name of the Faculty member: Mr.R.Deiva Nayagam

Name of the Topic: Multiple Access Techniques

Name of the Innovative Practice: Think Pair Share

Date & Time: 03.01.2020 & 15 Minutes

Description of Think Pair Share:

Think Pair Share is a co-operative discussion strategy that was developed by Frank Lyman and some of his colleagues in the early 1980's. It has three phases. During phase 1(Think) the instructor poses a question and allows the students time to ponder their response. During phase 2 (Pair) the student discuss their thoughts with their designated partner or tablemate. . During phase 3 (Share) the teacher will call upon the groups to have them tell the class what they discussed and think about that topic. This activity increases the collaborative learning skills of the students.

Goals(Learning Outcomes)

The students will be able to analyze and compare the various multiple access techniques.

Use of appropriate methods:

Justification for choosing the topic using Think Pair Share Activity:

This activity will helps the students to think individually about a topic or answer to a question and to share ideas with classmates and builds oral communication skills. It also helps focus attention and engage students in comprehending the reading material. Also the concept is very essential for everyone to know about the sharing of spectrum. In exam point of view, this concept is very important.

Effective Presentation

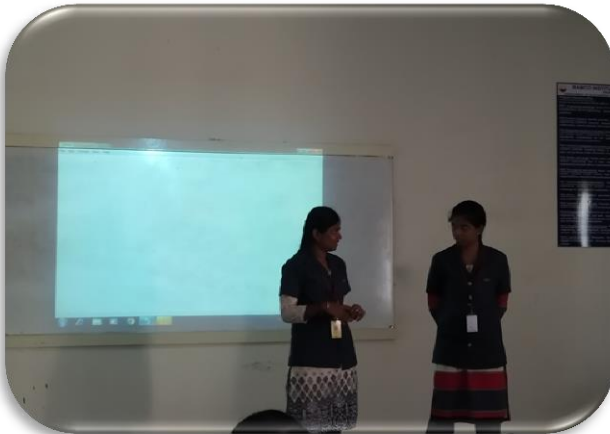
Details of Implementation

Think(T): During the first phase (T), I asked the students to think (individually) about the comparison among various multiple access techniques

- **Pair(P):** During the second phase (P), the students were asked to pair with their neighbours to discuss about what they think about the posed question
- **Share(S):** During the third phase (S), the students shared their thinking with their tablemate.

Execution

Multiple Access Techniques- Think Pair Share



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

The success of the activity was evaluated by asking this question in IAT I – Around 90% of students answered correctly. Since I conducted this activity, the collaborative learning skills of the students got increased. They actively participated in this activity.

Reflective Critique:

Benefits of conducting this activity

After Completion of this activity, students gave feedback that they got a clear idea about the spectrum sharing. They felt that it increases the cooperative learning.

Challenges

- Initially, the students not came up to involve in this activity voluntarily.
- It took a longer time than other strategies.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	2	3	2	1	1			2		2		2

CO2: The students will be able to apply the various cellular concepts like frequency reuse, channel assignments, handoff strategies etc., in mobile communication.

References:

1. Rappaport, T.S., "Wireless communications", Second Edition, Pearson Education, 2010
2. https://nptel.ac.in/content/storage2/nptel_data3/html/mhrd/ict/text/117102062/lec31.pdf.

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Degree, Semester& Branch: VI Semester B.E. ECE A

Course Code & Title: EC8652 Wireless Communication

Name of the Faculty member: Mr.R.Deiva Nayagam

Name of the Topic: Cellular Architecture Overview

Name of the Innovative Practice: Mindmap

Date & Time: 04.01.2020 & 15 Minutes

Description of Mindmap Activity:

Mind map is a tool that incorporates graphics, words, numbers, and color that represent and reflect a specific lesson or concept. It is a graphical way to represent ideas and concepts. Mind map is a visual thinking tool that helps structuring information, helping the students to better analyze, comprehend, synthesize, recall and generate new ideas. It is an easy way to brainstorm thoughts organically without worrying about order and structure. It allows the student to visually structure the ideas which is used to help with analysis and recall.

Goals(Learning Outcomes)

The students will be able to elaborate the cellular architecture concepts

Use of appropriate methods:

Justification for choosing the topic using Mindmap Activity:

The cellular architecture consists of various concepts and terms which is the very basic for wireless communication. So to make the students to enable meaningful learning and helps with memorization and retention. This activity makes the different concepts to understand easier.

Effective Presentation

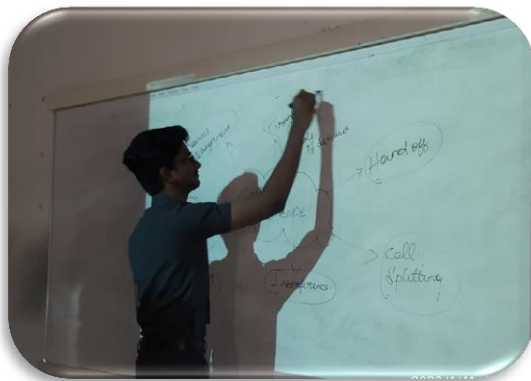
Details of Implementation

This mind map activity gives the graphical representation of words, concepts or pictures linked to and arranged around a given topic using a non-linear graphical layout which helps the students to build framework around a central concept.

I called the students one by one to come to the stage and write the term or concept related to this cellular architecture. So it was easy for them to recollect the points learnt in the classroom and they were able to view all the terms in a single activity.

Execution- Sample Images

Cellular Architecture- Mindmap



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

By conducting this mind map activity, the students were able to recall the topics they had learnt in the classroom and they were also able to reproduce it in the exam.

Reflective Critique:

Benefits of conducting this activity

After Completion of this activity, the students gave feedback about the better understanding of relationships and connections between ideas and concepts of cellular architecture

Challenges

NIL

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	2	3	2	1	1			2		2		2

CO2: The students will be able to apply the various cellular concepts like frequency reuse, channel assignments, handoff strategies etc., in mobile communication.

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2. <https://www.youtube.com/watch?v=kxLcwIMYmr0>
3. <https://www.youtube.com/watch?v=cJQZvxvDFug>